

VANZHILKYNAK, Russian Federation

WMO: 239660

Lat: 60.350N

Lon: 84.083E

Elev: 325

StdP: 14.52

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.6	-39.8	-48.6	0.3	-44.8	-43.3	0.5	-39.2	10.2	10.2	9.2	8.4	0.3	300	0.638

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.7	83.7	66.8	80.4	65.3	76.8	63.6	69.1	79.7	67.0	76.8	65.1	74.2	4.6	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.5	95.3	73.5	63.4	88.5	71.4	61.5	82.7	69.5	33.5	79.8	31.8	77.1	30.3	74.4	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
10.0	8.7	7.5	-48.9	88.4	6.9	2.6	-53.8	90.2	-57.9	91.7	-61.7	93.2	-66.8	95.1	
			DB												
			WB	-48.2	72.7	6.1	2.7	-52.6	74.6	-56.2	76.2	-59.6	77.7	-64.1	79.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.1	-8.8	-1.3	15.8	29.2	43.0	58.7	63.2	56.6	44.8	30.2	7.0	-3.6
	DBStd	27.76	17.15	15.27	13.16	11.02	10.36	9.11	6.62	6.87	8.07	10.41	17.92	18.48
	HDD50	9025	1822	1438	1061	623	271	28	1	18	197	616	1290	1660
	HDD65	13592	2287	1858	1526	1073	683	227	115	272	606	1080	1740	2125
	CDD50	1017	0	0	0	0	54	288	409	223	42	1	0	0
	CDD65	110	0	0	0	0	2	37	58	12	1	0	0	0
	CDH74	1335	0	0	0	0	61	477	652	138	7	0	0	0
	CDH80	310	0	0	0	0	13	119	154	24	0	0	0	0
Wind	WSAvg	2.7	2.0	2.3	2.9	3.1	3.3	2.8	2.5	2.4	2.6	2.9	2.6	2.3
Precipitation	PrecAvg	24.6	1.2	1.0	1.1	1.3	2.0	3.0	3.3	3.0	2.6	2.3	2.0	1.6
	PrecMax	30.8	2.6	2.4	2.7	2.5	4.5	6.0	7.4	6.1	5.4	4.2	3.3	3.1
	PrecMin	16.1	0.1	0.3	0.2	0.1	0.8	1.4	0.4	0.1	0.7	0.7	0.9	0.5
	PrecStd	4.1	0.6	0.4	0.6	0.6	0.9	1.1	1.8	1.5	1.1	0.7	0.7	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.2	32.0	45.7	60.8	81.2	87.4	88.3	83.6	74.2	57.0	37.7	31.5
		MCWB	29.3	30.2	38.1	47.9	63.4	68.0	70.3	68.3	61.0	50.0	35.8	30.6
	2%	DB	24.7	27.8	40.4	55.0	73.2	83.7	84.1	77.9	67.3	51.2	34.3	28.2
		MCWB	23.7	26.2	33.9	45.4	57.6	65.8	67.8	64.8	57.5	46.0	33.4	27.3
	5%	DB	20.2	23.7	36.5	50.4	66.9	80.0	80.9	73.6	62.2	47.2	32.9	25.3
		MCWB	19.3	22.2	32.2	42.1	54.2	63.9	66.4	62.7	55.2	43.6	32.1	24.4
	10%	DB	15.4	19.4	33.2	46.0	60.7	75.7	77.5	69.7	57.7	43.2	30.8	21.3
		MCWB	14.6	18.0	30.2	39.1	51.0	62.3	64.8	61.0	52.8	40.8	29.9	20.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.5	30.7	38.9	50.6	63.8	70.4	73.4	69.6	62.2	50.6	36.2	30.8
		MCDB	29.7	31.6	45.2	59.1	79.2	82.0	85.6	81.0	71.7	55.4	37.1	31.4
	2%	WB	23.9	26.4	35.0	45.9	58.8	67.4	70.2	66.6	58.7	46.9	33.7	27.4
		MCDB	24.8	27.5	38.8	53.5	70.9	79.3	80.6	75.2	65.2	50.5	34.1	28.2
	5%	WB	19.4	22.3	32.6	42.8	55.2	65.5	68.1	64.2	55.8	43.9	32.3	24.5
		MCDB	20.2	23.6	36.0	49.9	64.9	76.8	77.4	70.9	61.3	46.7	33.0	25.3
	10%	WB	14.6	18.1	30.2	39.4	51.7	63.6	66.2	61.9	53.0	40.8	29.8	20.4
		MCDB	15.4	19.4	33.4	45.1	60.1	73.6	75.0	68.6	57.3	43.2	30.7	21.4
Mean Daily Temperature Range	5% DB	MDBR	15.1	18.3	21.3	20.1	19.1	20.1	19.7	17.9	15.1	11.3	13.4	13.6
		MCDBR	16.3	15.7	22.2	25.0	29.3	27.7	25.9	25.1	24.3	16.8	9.0	14.3
		MCWBR	16.1	14.7	17.0	17.0	16.3	12.2	12.0	13.5	15.0	11.9	9.2	14.2
	5% WB	MCDBR	16.6	15.8	19.8	23.4	26.7	23.9	22.5	21.4	21.8	15.2	9.4	14.5
		MCWBR	16.4	15.0	15.9	16.3	15.6	11.8	11.7	12.9	13.9	11.2	9.6	14.4
Clear-Sky Solar Irradiance	taub	0.265	0.301	0.347	0.369	0.383	0.400	0.443	0.410	0.359	0.326	0.270	0.255	
	taud	1.961	2.006	1.964	2.046	2.275	2.329	2.184	2.264	2.372	2.257	2.016	1.904	
	Ebn at Noon	141	208	236	258	266	263	246	245	240	203	138	92	
	Edh at Noon	16	27	39	43	37	36	41	35	26	21	14	10	
All-Sky Solar Radiation	RadAvg	141	431	900	1398	1541	1764	1595	1168	708	328	148	72	
	RadStd	15	36	61	139	193	164	167	136	95	35	31	8	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page